MIL-C-47002(MI)
19 April 1974
SUPERSEDING
MIS-14124
30 June 1964

MILITARY SPECIFICATION

COATING, POLYURETHANE, SINGLE COMPONENT SYSTEM

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a flexible polyurethane single component system encapsulating compound.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-202	Test Methods for Electronics and Electrical Component Parts

FSC 5970

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

American Society for Testing Materials

ASTM D 115	Varnishes Used for Electrical Insulation, Testing,
ASTM D 150	A-C Loss Characteristics and Dielectric Constant Permitivity of solid Electrical Insulating Materials, Test For,
ASTM D 257	D-C Resistance or Conductance of Insulating Materials, Tests For,

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pennsylvania 19103.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproduction sample (4.2) of the compound shall meet the requirements of this specification.

3.2 Material. The coating compound shall be a single component, air curing, polyurethane formulation capable of being fully cured within 720 hours.

3.3 Electrical properties. The compound shall meet the requirements specified in Table I at 25 plus or minus 3 degrees Celsius (C) (77 plus or minus 5.4 degrees Fahrenheit (F)) when tested according to the applicable test in 4.6.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

Table I.

Electrical Properties

Property	Values	Applicable Test
Dielectric Strength (volts/mil)	2500, minimum	4.6.1.1
Dielectric Constant (60 to 10^6 cps)	3.6, maximum	4.6.1.2
Dissipation Factor (60 to 10^6 cps)	0.005, maximum	4.6.1.3
Volume Resistivity (10^{13} ohm-cm)	4, minimum	4.6.1.4
Surface Resistivity (megohm)	7500, minimum	4.6.1.5

3.4 Physical properties. The material shall meet the requirements specified in 3.4.1 through 3.4.5 when tested according to the applicable test in 4.6.2.

3.4.1 Appearance and color. The compound shall be homogeneous and free from lumps and coarse particles. A skin is permissable but shall be removed and discarded before use. The compound shall be yellowish in color.

3.4.2 Fungus resistance. The material shall show no evidence of fungus growth.

3.4.3 Shelf life. The shelf life of the material in unopened containers shall be a minimum of 6 months from date of manufacture when stored at temperature less than 27 degrees C (80.6 degrees F).

3.4.4 Moisture resistance. The moisture resistance shall be a minimum of 100,000 megohms.

3.4.5 Viscosity. The viscosity of the material shall be in the range of 4,000 to 4,300 centipoises.

3.5 Workmanship. The workmanship shall be such as to insure a product which is uniform, in conformance with this specification, and properly packaged and packed. The compound shall be free of dirt,

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

foreign material, or other contamination. Conatiners shall be clean, uniformly filled, well sealed, and legibly marked.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. The preproduction sample shall be subjected to all examinations and tests specified herein. Unless otherwise specified, the Government will perform the examinations and tests for preproduction sample acceptance at the contractors plant. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Classification of examinations and tests. The examination and testing of material shall be classified as follows:

a. Preproduction tests (4.4).

b. Quality conformance tests (4.5).

4.4 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

4.5 Quality conformance tests. Quality conformance tests for acceptance of material shall consist of the following examinations and tests:

a. Appearance and color (4.6.2.1).

b. Viscosity (4.6.2.5).

4.5.1 Lot size and sampling.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

4.5.1.1 Lot size. Lot size shall consist of all the material submitted for acceptance at the same time, which has been prepared by the same company without change in materials or processes in one continous period of operation.

4.5.1.2 Sampling. Unless otherwise specified (see 6.2), sampling shall be in accordance with MIL-STD-105.

4.6 Test methods.

4.6.1 Electrical property tests.

4.6.1.1 Dielectric strength. The dielectric strength shall be determined in accordance with ASTM D 115.

4.6.1.2 Dielectric constant. The dielectric constant shall be determined in accordance with ASTM D 150.

4.6.1.3 Dissipation factor. The dissipation factor shall be determined in accordance with ASTM D 150.

4.6.1.4 Volume resistivity. The volume resistivity shall be determined in accordance with ASTM D 257.

4.6.1.5 Surface resistivity. The surface resistivity shall be determined in accordance with ASTM D 257.

4.6.2 Physical property tests.

4.6.2.1 Appearance and color. The appearance and color shall be determined visually.

4.6.2.2 Fungus resistance. The material shall be tested for fungus resistance in accordance with MIL-E-5272.

4.6.2.3 Shelf life. The compound shall be tested for conformance with all the requirements specified herein after completion of storage period. The manufacturer's certificate of compliance with shelf life requirement (3.4.3) may be accepted during the storage period.

4.6.2.4 Moisture resistance. The moisture resistance shall be determined in accordance with MIL-STD-202, Method 106, after completing ten 24 hour cycles under polarizing voltages.

4.6.2.5 Viscosity. The viscosity of the material shall be determined in accordance with ASTM D 115 using a Brookfield viscometer, Model RVF, with a Number 3 spindle at 15 revolutions per minute.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified by the procuring activity, the compound may be furnished in tin plate cans or pails with metal lids. Containers shall be non-reactive to the material. Size of the contained shall be specified by the procuring activity. The material shall be prepared for shipment in such manner that damage during a shipment and storage is prevented.

5.2 Marking. Each container of material shall be legibly and durably marked in accordance with MIL-STD-129. Marking information shall include, but shall not necessarily be limited to, the following:

a. Manufacturers product designation.

b. Manufacturers name and address.

c. Lot or batch number.

d. Date of manufacture.

e. Shelf life.

f. Necessary precautionary notices.

g. Cure cycle.

6. NOTES

6.1 Intended use. The compound covered by this specification is intended for use in coating circuit boards by dipping, brushing or spraying. The compound, in fully cured condition, is effective in temperature range of minus 40 degrees C (minus 40 degrees F) to plus 121 degrees C (plus 249.8 degrees F).

6.2 Ordering data. Precurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required.

c. Where preproduction sample test shall be performed.

d. Sampling plan if other than specified (4.5.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

e. Any special marking requirements.

f. Container size.

6.3 Supersession data. This specification includes the requirements of MIS-14124, dated 30 June 1964.

Custodian: Preparing Activity:

in such manner that damage during a shipment and storage is prevented.

5.2 Marking. Each container of material shall be legibly and durably marked in accordance with MIL-STD-129. Marking information shall include, but shall not necessarily be limited to, the following:

a. Manufacturers product designation.

b. Manufacturers name and address.

c. Lot or batch number.

d. Date of manufacture.

e. Shelf life.

f. Necessary precautionary notices.

g. Cure cycle.

6. NOTES

6.1 Intended use. The compound covered by this specification is intended for use in coating circuit boards by dipping, brushing or spraying. The compound, in fully cured condition, is effective in temperature range of minus 40 degrees C (minus 40 degrees F) to plus 121 degrees C (plus 249.8 degrees F).

6.2 Ordering data. Precurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required.

c. Where preproduction sample test shall be performed.

d. Sampling plan if other than specified (4.5.1.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Missile Command
ATTN: DRSMI-RSDS
Redstone Arsenal, AL 35898

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assistdvn... Downloaded: 2026-10-22T07:32Z
Check the source to verify that this is the current version before use.